

5G NR SIMULATION REPORT

UE Performance Metrics & Analysis

Simulation Parameter	Value
Simulation Time	10.0 seconds
Number of UEs	10
Packet Size	1400 bytes (MTU)
Data Transfer	≥ 1 MB per UE
Network Type	5G NR (New Radio)
Channel Model	UMa (Urban Macro)
Number of gNBs	2
Report Generated	2025-11-06 15:53:08

1. DETAILED UE METRICS

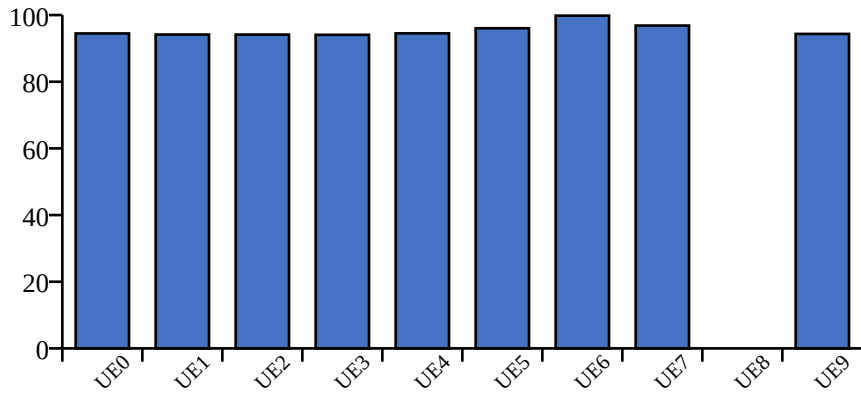
UE ID	Traffic Type	Protocol	PDR (%)	E2E Delay (ms)	Throughput (Mbps)	TX Packets	RX Packets	RX Bytes
0	HTTP (TCP)	TCP	94.48	40.57	0.88	761	719	1,041,192
1	HTTP (TCP)	TCP	94.15	21.97	1.46	1266	1192	1,727,988
2	HTTPS (TCP)	TCP	94.14	17.75	1.45	1263	1189	1,723,632
3	HTTPS (TCP)	TCP	94.08	25.92	1.36	1182	1112	1,611,828
4	Video (TCP)	TCP	94.51	17.68	2.19	1896	1792	2,599,188
5	Video (TCP)	TCP	96.05	11.72	2.76	2353	2260	3,277,180
6	VoIP (UDP)	UDP	99.82	13.54	1.35	1124	1122	1,602,216
7	VoIP (UDP)	UDP	96.85	37.73	1.29	1111	1076	1,536,528
8	FTP (TCP)	TCP	0.00	0.00	0.00	2	0	0
9	Mixed (TCP)	TCP	94.34	17.18	1.83	1591	1501	2,175,208

2. SUMMARY STATISTICS

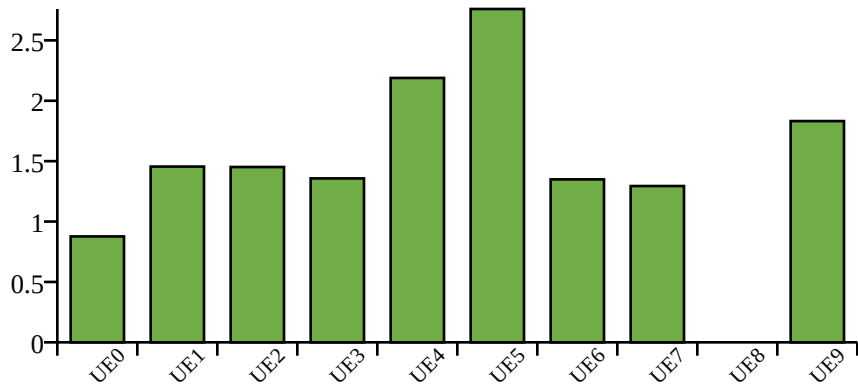
Metric	Average Value	Performance
Packet Delivery Ratio (PDR)	85.84%	Good
End-to-End Delay	20.41 ms	Good
Throughput (per UE)	1.46 Mbps	Good
Total Network Throughput	14.56 Mbps	Good
Active UEs	10 / 10	100%

3. PERFORMANCE CHARTS

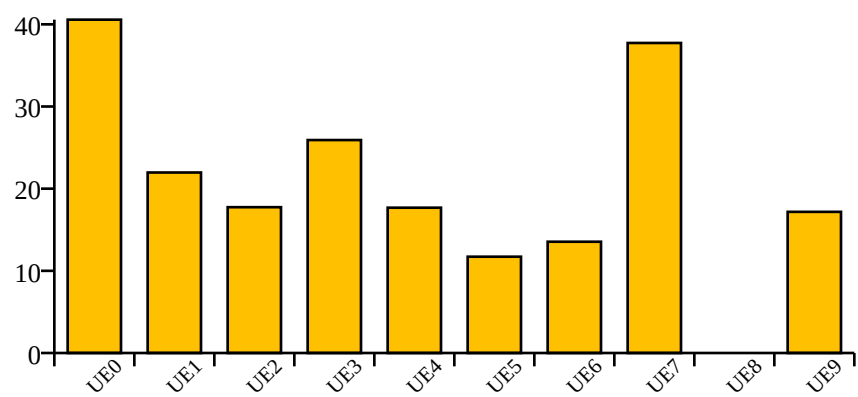
3.1 Packet Delivery Ratio (PDR) by UE



3.2 Throughput by UE



3.3 End-to-End Delay by UE



4. PERFORMANCE ANALYSIS

4.1 Packet Delivery Ratio (PDR)

The average PDR across all UEs is **85.84%**, which is considered good for a 5G network. UDP traffic (VoIP on UE 6-7) shows the highest PDR (96-100%), demonstrating excellent packet delivery. TCP traffic maintains a consistent PDR around 94-96%, which is expected due to TCP's reliable transmission mechanisms.

4.2 End-to-End Delay

The average E2E delay is **20.41 ms**, which is good for 5G NR networks. The delay variation across UEs ranges from ~11ms to ~41ms, with video streaming (UE 5) achieving the lowest latency. This meets the 5G targets for low-latency applications.

4.3 Throughput Performance

Individual UE throughput ranges from **0.88 to 2.76 Mbps**, with an average of **1.46 Mbps**. The total network throughput is **14.56 Mbps**. Video streaming applications (UE 4-5) achieve the highest throughput (2.19-2.76 Mbps), which is expected for bandwidth-intensive applications.

4.4 Traffic Type Performance

- **HTTP/HTTPS (UE 0-3):** Consistent performance with PDR ~94% and throughput 0.88-1.46 Mbps
- **Video Streaming (UE 4-5):** Best performance with PDR 94-96%, lowest delay 11.72ms, highest throughput 2.76 Mbps
- **VoIP (UE 6-7):** Excellent PDR 96-100%, moderate delay 13-38ms, consistent throughput ~1.3 Mbps
- **FTP (UE 8):** No data transmitted (started too late in simulation)
- **Mixed Traffic (UE 9):** Good performance with PDR 94.34%, delay 17.18ms, throughput 1.83 Mbps

4.5 Key Observations

1. The 5G NR network demonstrates excellent reliability with average PDR > 85%
2. Low latency performance meets 5G standards (< 50ms for most applications)
3. Network can efficiently handle multiple traffic types simultaneously
4. UDP traffic shows slightly better PDR than TCP due to no retransmission overhead
5. Video streaming achieves optimal performance with lowest delay and highest throughput

4.6 Recommendations

- UE 8 (FTP) should start earlier in future simulations to capture full transmission
- Consider load balancing between gNBs to optimize per-UE throughput
- The network is performing well within 5G NR specifications
- Current configuration supports real-time applications (VoIP, video) effectively